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Comparative Spatial Analysis Reveals Important Structural Similarities Between Ancient Angkor and Late Anglo-Saxon Hampshire

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Abstract

Angkor (800–1435 CE) is considered the type site for “low-density agrarian urbanism”, a loosely defined archaeological label used to describe settlements with urban traits but dispersed spatial structure. Angkor’s abandonment from the fifteenth century has fueled concerns that low-density cities, now common globally, may be inherently unstable. The spatial distinctiveness of Angkor relative to other pre-modern rural–urban systems, however, has never been quantitatively tested. Using ChronoCluster, a Python package for novel spatiotemporal archaeological analysis, we compared Angkor’s spatial structure to that of Anglo-Saxon Hampshire as recorded in the Domesday survey (mid-eleventh century CE), a rural–urban system not described as low-density agrarian urbanism. Surprisingly, both cases showed similar multiscale spatial clustering. Despite their different appearances, they may reflect scaled variants of the same underlying urban–rural dynamics. This targeted comparison between the “low-density” type site and a non-low-density comparator challenges the idea that low-density cities like Angkor form a distinct type, raises questions about what drives variation in urban form across time and space, and opens new questions about the impact of spatial structure on urban sustainability.

Keywords Urban archaeology · Urbanism · Spatial analyses · Clustering

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Introduction

Urbanism is one of the most enduring and recurrent societal organizing principles in human history. For at least 5500 years, and likely much longer, societies around the world have independently developed places that concentrate heterogeneous human activities, creating hubs of economic, political, and social interaction (Smith M.L. 2019, Smith M.E. 2023). These places intensify and multiply human interaction, giving rise to “energized crowding”, with urban spaces catalyzing interactions into social, cultural, technological, and economic innovations (Smith, 2023). Urbanism in this sense refers not merely to the presence of cities or their characteristic traits (*e.g.* Childe, 1950) but to the patterned organization of social relations that arises from the concentration and intensification of human interaction. Just as feudalism or capitalism are more than descriptions of political and economic conditions (Bloch & Koziol, 2014; Wood, 1999), urbanism is more than just a pattern of settlement or a general label for “city stuff”. Rather, it is a mode of social organization that reflects, shapes, and reinforces the social and economic relationships among people, emerging from and affecting individual and collective behaviour (Jacobs, 1992; Weber, 1958; Wirth, 1938). Many of these settlements are easily recognizable in the archaeological record because they frequently resemble the familiar cities of today in terms of functions and traits, like possessing public spaces, formal street networks, and public (non-domestic) architecture (Smith M.L. 2019, Smith M.E. 2023). Many also have high building densities relative to their surroundings and, in some parts of the world, walls that demarcate their boundaries. Others, however, seem to defy familiar models. The well-known Cambodian World Heritage site of Angkor, in particular, is the type site for what has been described as a highly distinctive form of premodern “low-density agrarian urbanism”, a classification that assumes fundamental differences from more compact urban forms (Fletcher, 2019; Fletcher et al., 2003).

Angkor was capital of the Angkorian World between the ninth and fifteenth centuries CE before being largely abandoned. Archaeological work at Angkor since the 1990s has revealed a vast 1000+ km² agro-urban, heavily modified landscape of temples, reservoirs, canals, roads, rice paddies, and dwellings (Evans et al., 2007; Fletcher et al., 2003). These findings evoked comparisons to modern forms of post-industrial urban sprawl thought not to exist in the premodern world (Fletcher, 2019). In the context of post-eighteenth century industrial urbanism, urban scholars had described sprawling, mixed-use regions with blurred urban–rural boundaries. Examples included metropolitan zones like Greater Vancouver (Hilmer, 1998), agro-industrial corridors like the U.S. Northeast Megalopolis (Lang et al., 2020), and Southeast Asian *desakota*—a portman-teau combining the Indonesian terms *desa*, meaning “village”, and *kota*, meaning “city” (Mcgee, 1991). However, thanks in part to a long-dominant conception of premodern cities as compact (often walled) settlements, these low-density mixed-use agro-urban regions were thought to be unique to the modern era, made possible largely by modern transportation technology (Fletcher, 2019). The accumulating archaeological evidence from Angkor challenged this assumption, and the

visual parallels with sprawling modern regions helped establish the “low-density agro-urban” model, now applied to the Classic Maya and other ancient centres in Asia and the Amazon Basin (Lucero et al., 2015; Prümers et al., 2022; Roberts et al., 2024). Alarmingly, scholars have recently pointed out that nearly all of premodern low-density cities we know of were abandoned or significantly depopulated (Fletcher, 2019; Lucero et al., 2015). These trajectories have led scholars (e.g. Fletcher, 2019) to warn that perhaps low-density urbanism is inherently unstable, and since post-industrial cities have actually been de-densifying around the world, there may be important consequences for the future of modern low-density urbanism.

Angkor, however, has never been quantitatively compared to another urban–rural settlement region. We addressed this gap by conducting the first comparative quantitative spatial analysis involving Angkor, focusing on the multi-scalar distribution of its community temples. These temples played a central role in religious life, political economy, agricultural organization, and local production (Fletcher et al., 2008; Hawken & Klassen, 2023; Klassen & Evans, 2020). More than places of worship, they were hubs for economic redistribution, labour mobilization, royal administration, dispute resolution, and power negotiation, reflecting a deeply integrated system of state control and sacred authority and tensions between local elites and royalty (Lustig, 2011; Lustig et al., 2007). Their role in sustaining the Angkorian system has been widely documented, particularly through rice and labour contributions to central administration (Lustig & Lustig, 2015). Importantly, the temples and their distribution in space played a key role in Angkor’s categorization as a low-density agro-urban complex (Fletcher et al., 2003).

For comparison, we examined Late Anglo-Saxon Hampshire, centred on Winchester. The Late Anglo-Saxon period refers to roughly the last two centuries before the Norman conquest of England in 1066 CE (Hamerow, 2014). We selected this contemporaneous comparator because sufficient data were available online and, importantly, because the Late Anglo-Saxon rural–urban settlement system was structured around manorial estates, which played roles similar to Angkor’s temples (see Table 1). Hampshire’s agro-urban system, however, was organized around a centralized market town embedded within a broader agrarian landscape and multi-tiered settlement hierarchy that most scholars view as compact or nucleated rather than low-density (Astill, 1991; Hamerow, 2014; Reynolds, 1982; Speed, 2014). While there is ongoing debate about the degree to which Late Anglo-Saxon and Early Norman England can be characterized as “fully urban” (Speed, 2014) by the time of the Domesday survey in 1066, Winchester was a thriving centre (Lavelle et al., 2021). Its dense population, mostly inside a circuit wall, were engaged in craft production and trade rather than directly or solely in agriculture (Hinton, 2000; Lavelle et al., 2021). Beyond Winchester, the county contained hundreds of manorial estates recorded in the Domesday Survey, which was started in 1066 CE and completed in 1086 CE (Harvey, 2014). These estates, whether secular or church-owned, formed the backbone of economic and political organization in Late Anglo-Saxon England (Gibbs, 2023). They structured land tenure, resource extraction, and local governance (Campbell, 2000). They supported agricultural production,

Table 1 Constituent-level element comparison. These variables were summarized from a number of sources for both Medieval Europe (Campbell, 2013; Hamerow, 2014; Harvey, 2014) and Angkor (Hawken & Klassen, 2023; Klassen & Evans, 2020; Lustig, 2011; Lustig & Lustig, 2015)

Comparative aspect	Manorial estates (Medieval Europe)	Community temples (Angkor)
Primary function	Feudal economic and social organization	Religious worship, community integration, agricultural production
Key contribution to rulers	Military service, rents, and dues	Rice and labour contributions
Core role	Local production and redistribution; religious for church-owned estates	Local production and redistribution, along with ritual and spiritual services
Connection to state	Feudal hierarchy linking lords and vassals	Centralized administration under the king (degree of centralization varying through time)
Autonomy	Local judicial and administrative autonomy, local production and labour organization	Centralized integration into royal networks, local production and labour organization
Economic foundation	Mixed agriculture/livestock, markets	Rice agriculture, craft production, trade
Infrastructure supported	Castles, manorial courts, and local defense	Temples, irrigation systems, and state projects
Cultural context	Secular with some religious undertones; overtly religious for church-owned estates	Deeply rooted in Hindu-Buddhist traditions
Type of obligation	Military and economic	Religious and agricultural

military obligations, and judicial functions, acting as the key nodes in a hierarchical system of lords, vassals, and monarchs (Campbell, 2013).

Functionally, therefore, manorial estates parallel Angkor's community temples, enabling a bottom-up comparison of two urban traditions (Fig. 1). In both systems, these units constituted the basic nodes of subsistence production, local redistribution, and obligations to a centralized authority positioned at a much larger and distinctive centre within their networks. This structural and functional correspondence, combined with the availability of point-location and chronological data for both systems, makes Hampshire the best publicly available comparator for Angkor. Crucially, our approach focuses on the constituents of each system rather than pre-conceived "sites" or "cities". Doing so allows us to avoid the tautology that labels Angkor a single site/city and then restricts comparison only to, for example, Winchester simply because both happen to carry urban typological labels. Angkor's

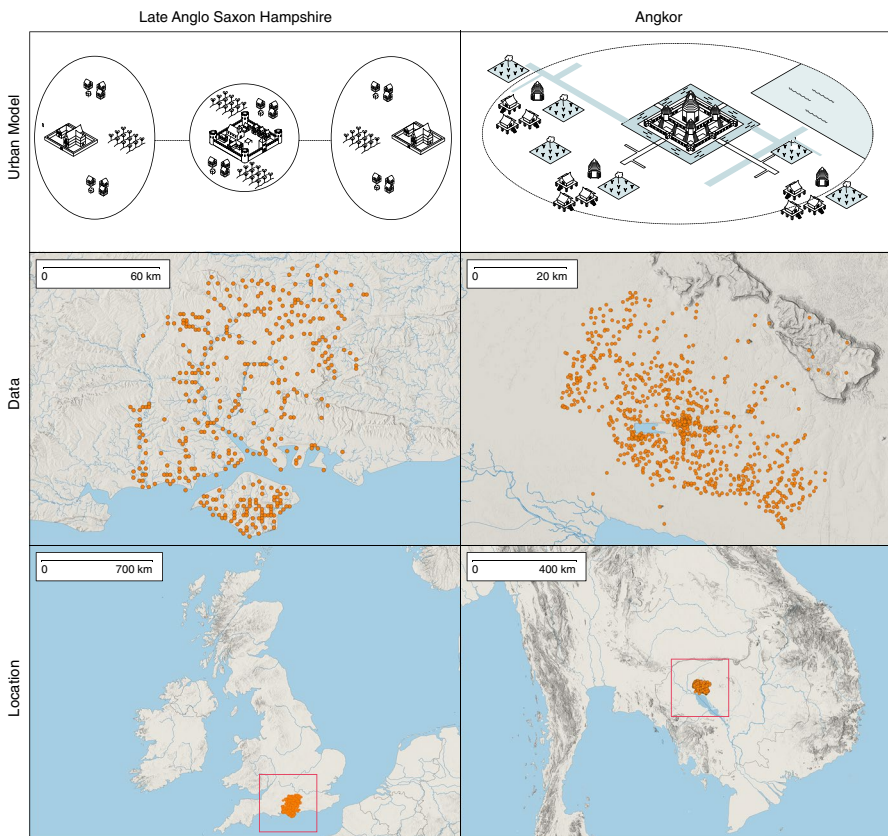


Fig. 1 Comparing urban models and data. Top row: our schematic representation of the two categorical labels circulating in the literature: the compact, nucleated, hierarchical settlement urbanism of Europe during the Middle Ages (left); and the low-density agro-urbanism of Angkor (right). The ovals indicate assumed boundaries in each case. Middle row: the constituent-level functional analogues used as data in the present analysis with manorial estates in Hampshire (left) and temples in Angkor (right). Bottom row: regional locations of both datasets

entire $1000 + \text{km}^2$ region containing temple communities and larger, walled centres (e.g. Angkor Thom) is generally considered a single urban complex, whereas Hampshire's manorial estates and its walled market-town centre are not. This deliberate asymmetry is the major axis of contrast that makes the comparison analytically useful: it allows us to probe whether the "low-density" label captures a real ontological distinction, or whether the type site of that label exhibits the same underlying spatial logic as a case that has not been described as low-density.

From our perspective, the high- and low-density categories are historically contingent and, clearly, relational rather than distinguished across a fixed density threshold. Nevertheless, it is useful to have some estimates to see that overall building and population densities in the two agro-urban regions were very different. In both cases, we can estimate building densities with minimum bounding convex hulls around the points used in each case, and we can estimate past population sizes using historically informed links between archaeologically or historically countable features or economic units (e.g. houses, parishes, community temples) and average numbers of individuals associated with the relevant unit. For Angkor, the coarse temple density across the study region is approximately 0.416 temples per km^2 (see SI for details). The population estimates depend on a number of variables, including the delineation of the urban–rural system in space—for our purposes and most recent scholarship, this area is the Greater Angkor Region covering $\sim 3000 \text{ km}^2$ —and the number of individuals per household and per community temple. The latter two variables are estimated from historical sources to be 5 and 100, respectively. Recent estimates, therefore, put Angkor's peak population density at approximately 233–300 persons per km^2 (Klassen et al., 2021). For Hampshire, we calculated the total manorial estate density to be approximately 0.086 estates per km^2 . We then modelled a distribution of population density estimates from historical data (see SI for details). Our estimated population density for Domesday Hampshire was approximately 14 per km^2 (95% credible interval 11.3–16.6), based on a Bayesian model propagating uncertainty in mean household size informed by Darby's reference range and three reconstructed literature estimates (Darby, 1986; La Poutre, 2023).

Alone, however, these density estimates say little about spatial logic or urban–rural processes and any arbitrary threshold could be shifted. Thus, density values, ironically, cannot be used to test the ontological force of the prevailing categorical distinctions between high- and low-density premodern urbanism. Therefore, we turned instead to a more appropriate tool: pairwise distance density (PDD) functions. PDDs express the frequency of interpoint distances in a point set and are the mathematical foundation of clustering statistics. They reveal both clustering and inhibition (i.e. spatial dispersion) in point processes at multiple spatial scales, revealing multiscale spatial logic behind point patterns. Using PDDs, we examined whether the multi-scalar clustering structure of Angkor's community temples differs from that of Hampshire's manorial estates, across "near" (short interpoint) and "far" (long interpoint) distances where "near" and "far" are relative to the local geographic context and spatial extent in each case.

We compared the PDD functions of the two cases with the *ChronoCluster* Python package (Carleton & Song, 2025). Using *ChronoCluster* tools, which account for chronological uncertainty, we identified statistically significant deviations in the

PDDs by subtracting out appropriate null models—one we call a baseline informed spatial expectation (BISE) and the other was complete spatial randomness (CSR), which we included for consistency with traditional point pattern analysis (PPA) in archaeology. This operation produced ΔPDD functions that revealed significant features, including clustering and repulsion, across interpoint distances, providing insight into the underlying spatial logic of settlement organization in each region and enabling clear, direct comparisons (Bevan & Conolly, 2006).

Our aim was to conduct a targeted comparative test of whether the spatial logic of the canonical “low-density” type site differs categorically from that of a well-documented agro-urban system that has not been described as low-density. We reasoned that if the ΔPDD s in the two cases revealed different spatial structures, the finding would reinforce a prevailing notion that Angkor represents a distinct kind of premodern urbanism in accordance with the low-density label (providing some support for the ontological force of the label). If, however, the results revealed similar underlying spatial structures, the finding would challenge the idea that Angkor’s low-density urbanism differed structurally from that of Late Anglo-Saxon Hampshire, suggesting that these two seemingly distinct urban–rural traditions represent variations on a common spatial logic. This would imply that Angkor may not represent a categorically distinct type of premodern urbanism, raising questions about the ontological distinctiveness of the low-density category.

Materials and Methods

Data and Study Design

Our analysis involved two sets of point data (Carleton, 2025). The first was derived from a published database of 1431 community temple locations in the Greater Angkor Region (Carleton et al., 2023; Klassen et al., 2018). Since only 105 of those temples have foundation dates estimated directly using epigraphy and/or art and architectural assessments, we followed the approach of Carleton *et al.* (2023) to estimate posterior foundation date distributions for the 1326 undated temples. Of those, a subset had spatial coordinates, resulting in a final sample size of 1286 temples. We used the modelled foundation date distributions of those 1286 temples and the historically defined end of the Angkorian period (1435 CE) to define spans and uncertainties for the temporal presence of each temple (see SI for details).

The second dataset consisted of 435 manorial estates located in Hampshire listed in the Domesday Book (Harvey, 2014). The book reports a land survey conducted by William I after conquering England in 1066 CE that was concluded in 1086 CE. The contents of the book were translated and digitized in 2015 at the University of Hull and the National Archives. We downloaded the raw data, including point locations, from the University of Hull website (Palmer et al., 2015). We then isolated the points listed as within the county of Hampshire and cleaned a couple instances of points well outside the county having been duplicated and mislabelled as belonging

to “Hampshire”. The temporal presence of the estates was taken to be 1066–1086 CE, effectively the time taken to complete the Domesday survey.

We then identified statistically significant clustering and dispersion across spatial scales in each point set and compared them. These analyses accounted for chronological uncertainty and potential variability (in the case of Angkor). Mathematical details and extended analyses are available in the SI. A full computational notebook (Python Jupyter Notebook) that can be used for replication is also available along with the necessary data on GitHub (<https://github.com/wccarleton/angkorclusters>) and archived with Zenodo (<https://doi.org/10.5281/zenodo.15371728>).

Statistical Analyses

PDDs characterize the frequency distribution of interpoint distances across all possible pairs of points in a given point set, enabling direct comparisons between point patterns across multiple spatial scales. A PDD is constructed by measuring the distance between every possible pair of points—temples in the case of Angkor and manorial estates in the case of Hampshire—and examining how frequently each distance occurs. The resulting smooth density curve shows which interpoint distances are more or less common within the dataset. Peaks in the curve indicate distances that occur relatively frequently among point pairs, while troughs indicate distances that occur less often or are largely absent. On their own, these peaks simply indicate modal interpoint distances; clustering or dispersion can only be inferred by comparing the empirical PDD with expectations under appropriate null models. In practical terms for the present study, the PDD summarizes the spatial spacing pattern of the institutions that structured economic, social, familial, political, and religious life in Hampshire and Angkor.

The PDD itself is an empirical summary of the point pattern, and many spatial statistics designed to detect clustering are derived from it. However, unlike derivatives such as Ripley’s *K*, the PDD directly represents the full distribution of pairwise distances and therefore captures clustering, dispersion, and broader structural features of the point pattern simultaneously across multiple spatial scales. Importantly, this approach does not require assuming complete spatial randomness (CSR) or a homogeneous Poisson process as a baseline for detecting clustering, nor does it require selecting arbitrary analysis radii (both of which are assumptions required by the derivative statistics). Instead, assessing the PDD directly allows the use of simulation-based null models that incorporate first-order spatial inhomogeneity (*e.g.* the overall shape and extent of the point pattern). For comparison with traditional archaeological point pattern analyses, we also include results using CSR as a null model (see SI). Detailed definitions of the PDD are provided in the SI.

To account for chronological uncertainty and potential change over time in archaeological point patterns, we used a new python package called ChronoCluster (<https://github.com/wccarleton/chronocluster>). The package adopts a spacetime perspective for modelling archaeological data, which means that both space and time are considered dimensions that make up a continuous spacetime volume (Maudlin, 2012)—see Fig. 2. A cornerstone of the package is the calculation of “inclusion

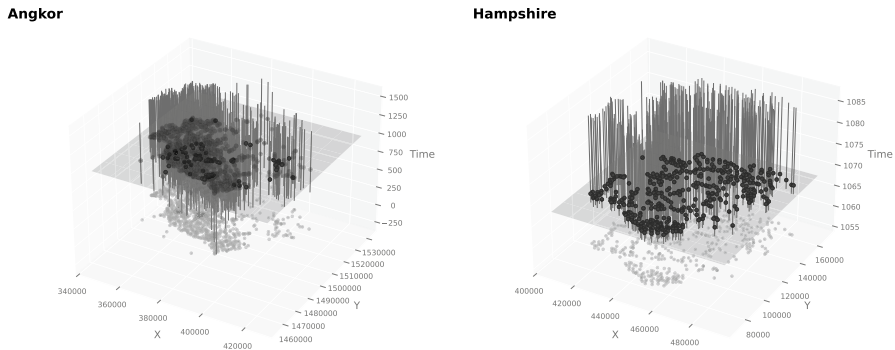


Fig. 2 Spacetime volumes of Angkor's temples (left) and Hampshire's manorial estates (right). The x and y -axes represent spatial coordinates (in this case in meters in the regions' respective projected UTM coordinate systems) while the z -axis represents time. Vertical pipes indicate the presence of a given point over time. The horizontal plane is an arbitrary slice of time and the black points are the points present in the pattern at the time indicated by the plane. The gray points on the floor of each plot represent the atemporal spatial locations of all points (*i.e.* the palimpsest of points on the landscape). Inclusion probabilities can be calculated for the points at any given time slice (horizontal plane) and then used for bootstrap sampling or directly as weights in spatial statistics

probabilities", which express the probability that a given event (*e.g.* a temple) was present in the relevant point pattern at a specified time. Formally, the inclusion probability of an event at time t is the probability that the event has started but not yet ended:

$$P_A(t) = P(T_{start} \leq t \text{ and } T_{end} \geq t)$$

Let $F_A(t)$ and $G_A(t)$ denote the cumulative distribution functions of the start and end times of event A , respectively. Under the assumption that start and end times are independent, this becomes:

$$P_A(t) = F_A(t)(1 - G_A(t))$$

These functions can represent any probabilistic chronological information, including calibrated radiocarbon dates or other chronometric estimates.

This formulation defines a continuous-time probabilistic occupancy function for each event, allowing ChronoCluster to represent both temporal dynamics (changes in the point pattern through time) and chronometric uncertainty in a unified way. Rather than assigning events to discrete time slices, each event contributes to the point pattern at time t in proportion to its inclusion probability.

In practice, we implemented this using ChronoCluster's bootstrap framework, which propagates uncertainty in event contemporaneity by sampling realizations of the point pattern at each time step. For a given time slice, points are probabilistically included according to their inclusion probabilities, generating an ensemble of plausible point configurations. In the case of Hampshire, the inclusion probabilities were estimated using fixed historical dates based on the known date of the Domesday survey, and for the Angkorian temples, we used a combination of epigraphic

and modelled (predicted) foundation dates (Carleton et al., 2023) as the start dates with uncertainties and a fixed historical end date for all temples corresponding to the canonical historical end of the Angkorian period (1431 CE). The modelled start dates for the temples were produced with a Bayesian model that used the subset of temples with epigraphic or art historically determined foundation dates as training data to predict missing foundation dates on the basis of correlations between known foundation dates and several spatial and architectural variables shown to have chronologically predictive utility (Carleton et al., 2023; Klassen et al., 2018). The resulting ensembles were then used to compute distributions of pairwise distance densities (PDDs) for each time slice, and to track their evolution through time. This approach avoided the need for arbitrary temporal binning and instead treats contemporaneity as a probabilistic property of events, preserving both uncertainty and temporal continuity.

As mentioned, the empirical PDDs were compared to two null models. These null models allowed us to estimate whether features of a given PDD (modes or troughs) could be considered “significant” in the sense that they differed substantially from what would be expected under the given null. The first null model, *complete spatial randomness* (CSR), represents a standard assumption in point pattern analysis (PPA) (Ripley, 1981), while the second, which we term *baseline informed spatial expectation* (BISE), incorporates first-order effects like anisotropy, point pattern size/variance, and location within an arbitrary study area (details are available in the SI). Consequently, the BISE better isolates finer-scale structure within the observed point patterns while remaining empirical and avoiding the need for full process modelling that would be entailed in spatial regression, or inhomogeneous Poisson models. Thus, the BISE is in line with recent developments in archaeological PPA whereby scholars have recognized the general inadequacy of CSR for identifying meaningful patterns in archaeological data (Bevan, 2020). It is also highly empirical and aligned with our aim of directly comparing multi-scale spatial structure rather than explaining each point pattern.

Like any null model, the BISE accounts for some sources of structure and not others. It does not, for instance, account for all local or second-order landscape constraints that could influence interpoint distances—e.g. the Barays at Angkor or strait between Hampshire and the Isle of Wight. With such large regions and many points throughout each, it is highly likely that those specific, localized constraints would have comparatively small effects on the distribution of interpoint distances described in the PDDs. Any modes in the PDDs specific to the corresponding dimensions of the constraining features (i.e. length/width of the Barays, and average width of the strait) would also represent the frequency of those same interpoint distances elsewhere in the two regions, and since there are many points in each case that are much closer to one another than the constraining distances, we would expect to not see troughs in the PDDs corresponding to distances smaller than the constraining dimensions. We reasoned that if we saw such features in the PDDs, then we would need to mask those features out of the null models in order to remove their effects.

With that in mind, we used deviations from these nulls as a measure of significance for features in the PDDs. Deviations of the empirical PDDs from null models (ΔPDD s) for each region were identified by simulating distributions of differences

in a point-wise manner across a series of interpoint distances. For a given interpoint distance, a deviation was considered significant if the inter-quantile range between the 5% and 95% quantiles of the ΔPDD distribution excluded zero (see SI for further details). This effectively created p -values that could be used to support inferences about statistical significance.

Results

Our results revealed key similarities between the two case study regions. The similar patterns were evident in PDD (Figs. 3 and 4) and ΔPDD model comparisons (Figs. 5 and 6), and visible even after accounting for known, quantifiable chronological uncertainty (also represented in Figs. 5 and 6). Temporal stability at Angkor from early in its temple construction boom—visible in a heatmap visualization (Fig. 5) time series of ΔPDD s (Figure S8)—permitted the direct comparison between Angkor (1000 CE) and Hampshire (effectively at 1086 CE) central to the present study. Importantly, that comparison revealed striking parallels in the spatial logic underlying their respective settlement structures.

The comparison between the ΔPDD s from Angkor (1000 CE) and Hampshire (1066–86 CE), shown in Fig. 7, revealed striking similarities. In each case, the function increases rapidly at the low-distance end of the domain and

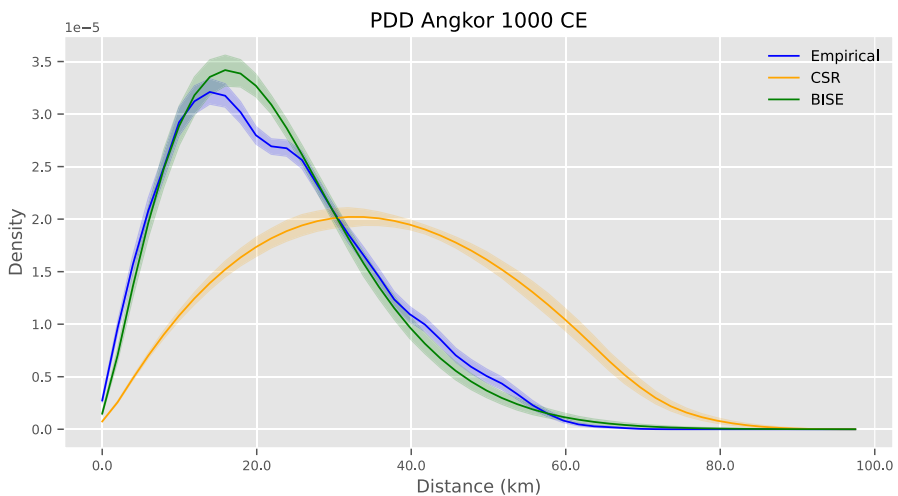


Fig. 3 Pairwise distance density (PDD) functions for Angkor’s temples at 1000 CE (one representative time slice). The densities each reflect all 826,255 interpoint distances (excluding self-comparisons) that could be created with the temples data. The densities shown reflect the relative frequencies of interpoint distances (x -axis) in the dataset including the actual observed density function (blue), and the same points with coordinates drawn at random from a CSR (green) and BISE (red) null models. The two nulls indicate the expected distribution of the temples if their coordinates (and resulting multi-scalar spatial structure) were determined by the relevant null process. The ribbons indicate both spatial and temporal uncertainties arising from inclusion probabilities (uncertain foundation dates) and in the case of the null models different equally probable randomized coordinates for each temple location

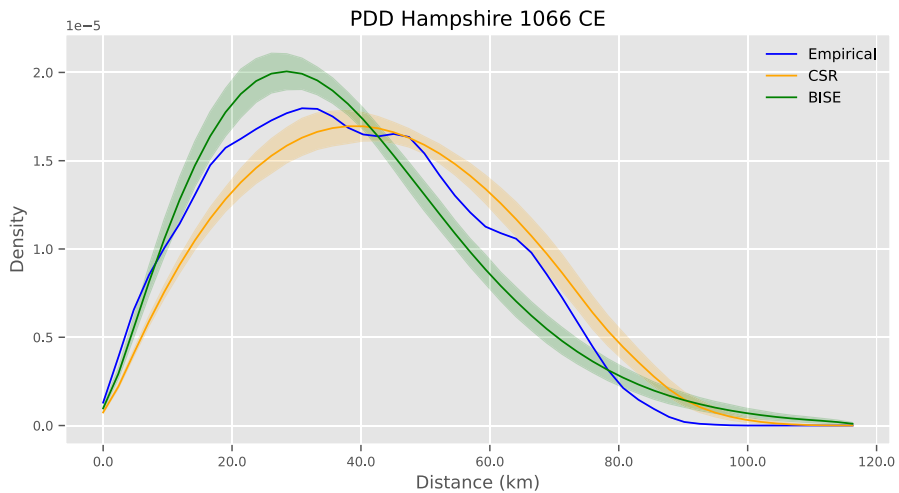


Fig. 4 Pairwise distance density (PDD) functions for Hampshire's manorial estates at 1086 CE (one representative time slice). The densities each reflect all 94,395 interpoint distances (excluding self-comparisons) that could be created with the manorial estates data. The densities shown reflect the relative frequencies of interpoint distances (x -axis) in the dataset including the actual observed density function (blue), and the same points with coordinates drawn at random from CSR (green) and BISE (red) null models. The two nulls indicate the expected distribution of the temples if their coordinates (and resulting multi-scalar spatial structure) were determined by the relevant null process. The ribbons indicate spatial uncertainties arising from different equally probable randomized coordinates for each estate location under a null model. Given that we are working with fixed historical data, chronological uncertainty was not relevant for the Hampshire data, which is why the blue empirical density has no uncertainty envelope (all manorial estates were considered to have been present at the given time with no uncertainty about inclusion probabilities)

then declines, forming a first discrete mode. Then, the function declines into a large negative trough, before rebounding into the positive space and following a trajectory that creates a second, wider mode. Lastly, both functions come down again off the second mode, trailing to effectively zero for the rest of the function domain. In neither case was there any indication of PDD features related to specific local constraints like the Barays, canals, rivers, or strait (in the case of Hampshire), which led us to conclude that for our purposes, it was not necessary to mask those features.

The damped oscillation of the PDDs indicates the presence of two statistically significant modes in the PDDs of both point sets. The first mode is much more distinct, implying a shorter-distance clustering where we would observe higher densities of points (estates in the case of Hampshire and temples at Angkor). The second mode, being much wider and more gradual, implies some large-scale clustering beyond what would be expected under the BISE, perhaps a second-order clustering of the more discrete, smaller-scale clusters responsible for the first mode. But, this second mode is much less distinct than the first, which suggests it would be harder to identify visually. Nevertheless, in both cases the bimodal, damped oscillator structure implies at least two scales of clustering are

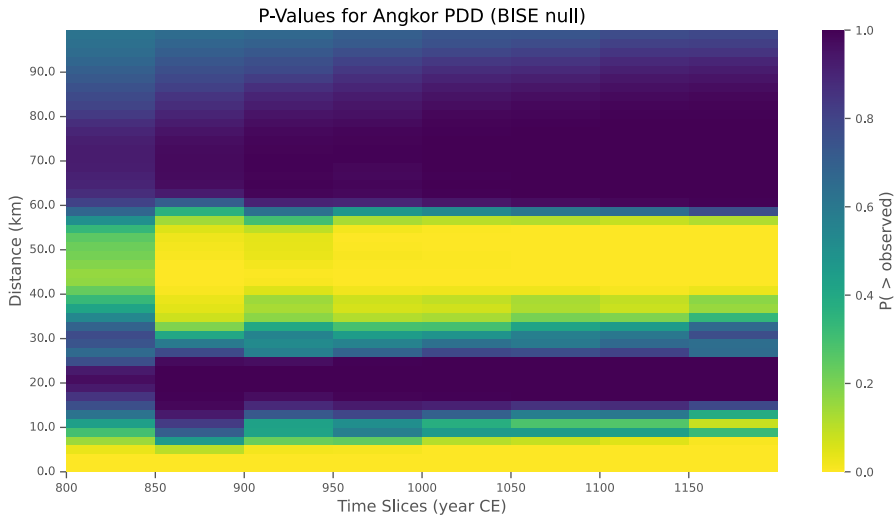


Fig. 5 Heatmap of p -values indicating the probability that the PDD for Angkor was significantly different from the BISE null model. Hotter regions indicate statistically significant spatial structure in the point pattern that differs from the null expectation. The p -values (and underlying PDDs) at each time slice are based on 826,255 interpoint distances

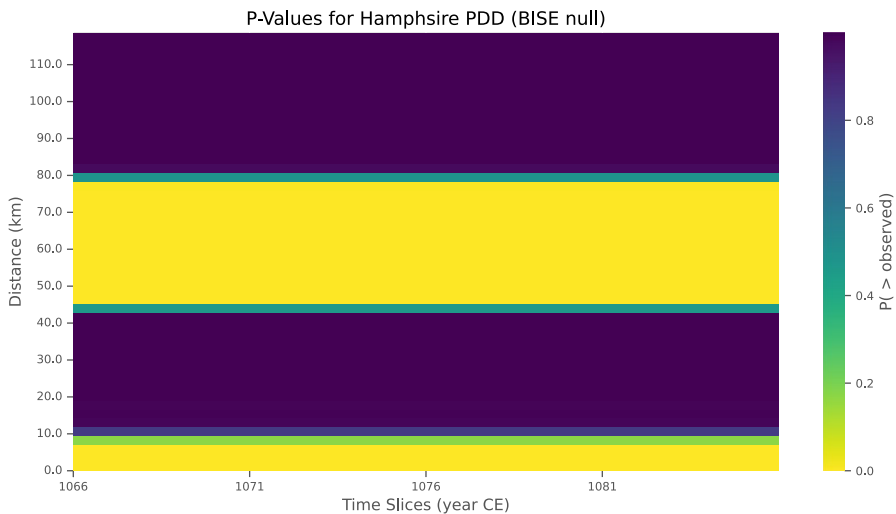


Fig. 6 Heatmap of p -values indicating the probability that the PDD for Hampshire was significantly different from the BISE null model. Hotter regions indicate statistically significant spatial structure in the point pattern that differs from the null expectation. The p -values (and underlying PDDs) at each time slice are based on 94,395 interpoint distances

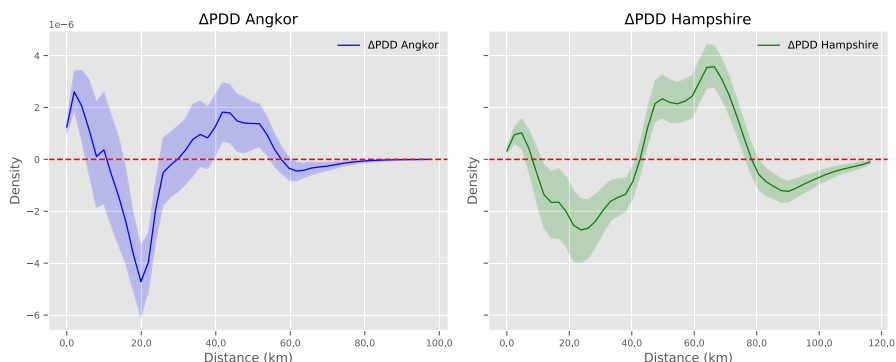


Fig. 7 ΔPDD s for Angkor (left) and Hampshire (right) based on the BISE null model. The uncertainty ribbons represent both chronological uncertainty in start and end dates and spatial variability in the underlying BISE null. The envelopes represent the point-wise 5–95% bootstrapped quantiles from 500 iterations. The horizontal dashed lines represent zero—no difference between the null and the empirical point pattern at a given interpoint distance. For Angkor, the ΔPDD s is based on 826,255 interpoint distances, while for Hampshire it is based on 94,395 interpoint distances

observable (under the BISE null model) and the functions are very similar across interpoint distances given a consistent null model used as a basis for comparison. The implication is that in both cases, rural–urban life was spatially hierarchical in the sense that local communities, like groups of neighbouring community temples or nearby manorial estates, were situated within larger clusters of those groups.

In fact, the wave-like pattern appeared to us so similar between the two cases that we investigated further (see SI for details). It seemed clear that one could stretch or compress either of the ΔPDD functions along its x -axis and the main peaks and troughs of the waves in the two cases would align. We rescaled Angkor's ΔPDD along its x -axis, which effectively means scaling all of the interpoint distances such that the longest distance across the settlement areas would be made equal—this would be analogous to expanding or contracting the diameter of one circle to match another. When we stretched Angkor's ΔPDD in this way, corresponding to a factor of around 2, its peaks and troughs broadly align with Hampshire's (Fig. 8)—we confirmed the scale factor in a supplementary analysis (see SI) in which we quantitatively compared the first mode locations while accounting for uncertainties (see the distribution in Fig. 8). This convergence suggests that rather than representing two distinct urban traditions, these settlements may be scaled manifestations of a shared underlying spatial logic.

Discussion

If Angkor was a fundamentally different type of urbanism, we would expect its spatial structure to be starkly divergent from a reasonable comparator that has not been given the same categorical label. Yet, we found some compelling similarities

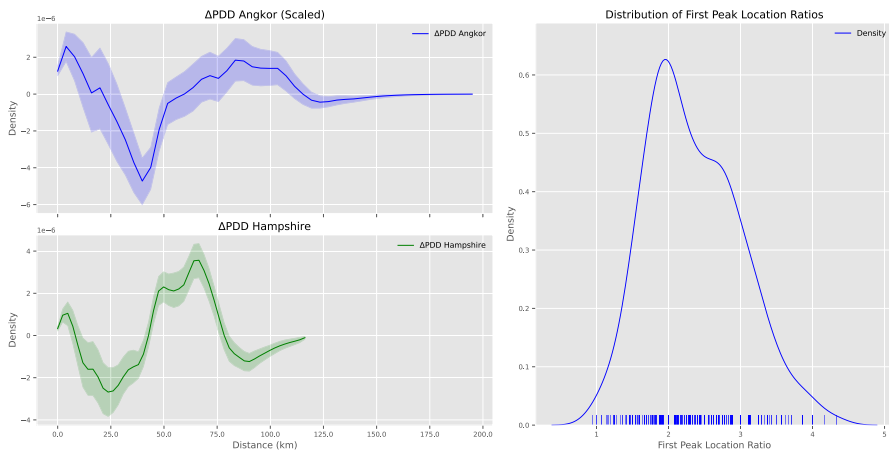


Fig. 8 ΔPDD s for Angkor and Hampshire based on the BISE null model (left)—Angkor’s function has been stretched by a factor of 2, demonstrating the scaled similarity between the spatial logics of both regions. On the right, the distribution represents the ratio of first-peak locations of Hampshire’s to Angkor’s ΔPDD that accounts for chronological uncertainty and spatial variability in the BISE null derived from 500 bootstrap iterations

in multi-scalar spatial clustering of Angkor’s community temples and Hampshire’s manorial estates. In fact, we found that the clustering patterns appear to be scaled versions of one another, with a scaling factor of approximately 2 (*i.e.* Angkor’s clustering pattern looks remarkably like Hampshire’s only compressed by the scaling factor). This finding resonates with recent research on scaling in urban systems (*e.g.* Carleton et al., 2025; Ortman et al., 2020) and suggests that common constraints may have shaped the rural–urban systems of the Greater Angkor Region and Late Anglo-Saxon Hampshire.

From this pattern, we can infer that the institutional landscapes of both regions were organized into nested spatial groupings. In practical terms, this suggests that routine economic, social, and religious interactions were structured primarily within relatively local clusters of communities, while broader administrative and political relationships linked these clusters into larger regional networks. Existing interpretations of historical and archaeological evidence from both regions are broadly consistent with this interpretation (Faith, 1999; Hamerow, 2014; Hawken & Klassen, 2023; Hendrickson et al., 2023). A major difference, then, between the two cases involves the space between nodes and clusters. Angkor was decidedly denser than Hampshire, likely resulting in higher total interaction potential (more face-to-face meetings between more people) during routine daily activities and more frequent interaction between community temple groups than between Hampshire’s manorial estate groups. And although the cultural meanings of these interactions differed between the Khmer and Late Anglo-Saxon contexts, the spatial structure organizing these relationships appears comparable in both systems.

This similarity has two important implications. First, the now widely accepted exceptionalism of Angkor relative to other premodern urban systems may need to

be reconsidered in light of our comparison. In particular, the idea that its entire 1000+ km² region represents a special “low-density” agro-urban complex that is categorically different than nucleated or compact urbanism should be reconsidered. Since its introduction to the archaeological literature, the low-density label has been applied beyond Angkor to Maya cities, to ancient Sri Lankan centres, and to Amazonian settlements that have only recently been mapped with lidar (Fletcher, 2019; Prümers et al., 2022). Our finding, however, calls into question the distinctiveness of the type site for the low-density label, which raises questions about the validity of the categorical distinction itself. While it is clearly the case that cities, past and present, vary in terms of density and form (among cities and within individual cities), and research into this variation is key (Fletcher, 1995, 2019), whether there are or have been categorically distinct urban trajectories matters tremendously for our understanding of urbanism and the way it arranges our societies, economies, relationships, landscapes, and cultures.

Second, if our findings are replicated, apparent differences in urban form should not immediately be taken as evidence of distinct types of urbanism. The dominant, top-down approach to defining and discussing premodern urban systems (*e.g.* low-density versus nucleated) has relied on assumed categories, treating features such as walls and density estimates as definitive markers that can be used to distinguish types of urbanism. Yet, in doing so, it risks what might be called “begging the category”. We see this as a form of circular reasoning in which cities are classified based on predefined expectations, and those classifications are then used to justify the very distinctions they presume. Instead of first identifying what actually structured life within a given settlement area and testing whether types emerge from lower-level patterns, top-down approaches begin by assigning a category label, like “compact” or “low-density”, drawing a boundary, and estimating traits (*e.g.* densities) based on those initial assumptions. And if, for example, the absence of a wall both determines the label and affects how we define spatial boundaries for estimating densities, then using the resulting density estimate as support for the label is circular, especially if structures, habitation, and activity continued beyond the boundary (Hammer & Di Michele, 2023; Yang et al., 2024). That is, the system may be low-density merely because we have defined it that way and drew a large enough boundary, rather than because something about its constituents distinguishes it categorically from other systems.

For our study, we used an alternative, lower-level approach. We began by identifying the constituent structures at Angkor whose mapped distributions have been central to its visual classification as “low-density”. We then examine their social and economic roles, identify close functional analogues in another well-documented case—Hampshire’s manorial estates—and compare the two systems directly. This comparative low-level approach revealed that in these two cases rather than discrete urban types, the urban–rural landscapes of Hampshire and Angkor exhibit variations of a shared underlying spatial and functional logic. If such structures emerge repeatedly across different regions and time periods, then the notion of low-density urbanism as a distinct category may not reflect actual urban diversity, but rather an artifact of circular classification. This is not to call into question the significance of insights into density as an important variable when considering urban formation, adaptation

and resilience (Fletcher, 1995, 2019; Fletcher et al., 2003), but rather to study this as a continuum and process rather than a taxonomy of “types”.

While revealing, our study has two key limitations. First, we compared only two cases directly (although supplementary analyses revealed similar looking patterns in most Domesday counties—see SI and Figures S9 through S19). It remains possible that other ancient cities currently identified as distinctive low-density types—such as Classic Maya centres or recently mapped Amazonian settlements—would continue to appear distinct. However, our findings demonstrate, at a minimum, that Angkor, a commonly referenced type site for the low-density model, is more similar than previously recognized to a case that has not been labelled low-density. Importantly, as we argued, there are sufficient socioeconomic and political parallels with respect to the meaning of the individual points in both cases to justify a direct comparison. This challenges the validity of the categorical label distinctions, but we cannot yet determine whether a distinctive low-density category exists, or existed at some point, globally.

Second, our study was constrained by the available data. For Hampshire, we primarily relied on the Domesday Survey, which provides only a snapshot of the urban–rural system as it existed in the Late Anglo-Saxon period. For Angkor, our chronology is based largely on modelled dates for temple construction and a single historically normative date for abandonment (the conventional endpoint of the Angkorian period in the mid-fifteenth century). To fully capture the temporality of urban development, it would be ideal to have higher-resolution chronological data for Angkor and a more temporally variable point pattern for the comparator. It is possible, for instance, that the urban–rural system of Late Anglo-Saxon Hampshire changed significantly after the period for which we have data. Still, after accounting for chronometric uncertainties—including modelled uncertainties for the Angkor point set—our results indicate that the spatial structures observed in both cases persisted for at least several centuries. Thus, at the point of comparison, the patterns we identified reflect real, long-standing urban forms rather than transient anomalies and the distinctiveness of putative low-density urban systems should be re-evaluated.

We identify two major avenues for future research. First, if our findings are replicated across cultures and time periods, the apparent dichotomy between compact and low-density urban forms may reflect positions on a continuum rather than categorical differences—an empirical question that only broader comparison can resolve. Once appropriate datasets have been compiled, our study could be repeated for point sets representing Classic Maya urbanism in Mesoamerica, Anuradhapura-period urbanism in Sri Lanka, and pre-Colonial Amazonian urbanism, all of which have been classified as low-density urban forms (Fletcher, 2019). Likewise, other premodern cases often considered more “compact” examples (*e.g.* other European medieval urban regions (Nicholas, 2014), Chinese walled cities (Turnbull & Noon, 2009), ancient Greek and Roman centres (Zuiderhoek, 2016)) should be analyzed to determine whether these, too, fall somewhere on a scaling continuum or are indeed different in some spatial structural way from those classed as low-density.

The source of the observed scaling difference also warrants further investigation. As we noted, the scaling factor relating Angkor to Hampshire was about 2. This specific value may be significant. For instance, the ratio between the speed of an

ambling horse and that of a walking human is also roughly 2 (Bohannon & Williams Andrews, 2011; Stefánsdóttir et al., 2021; Wutke et al., 2016). In medieval England, specific horse breeds were preferred for a specific ambling gait, particularly among elites who needed to move efficiently between manorial estates (Jenély, 2015). In contrast, horses were not widely used at Angkor (Cremin, 2015) and have generally been uncommon in the tropics (De Jong & Prins, 2023), where most premodern low-density cities have so far been identified. Other draft animals, like elephants, were available at Angkor (depicted in bas reliefs on several temples and monumental structures and described in the first-hand thirteenth century account of Zhou Daguan (Zhou et al., 2007)), but have a normal walking speed comparable to humans (Hutchinson et al., 2006).

Another potential explanation that needs exploring involves the relative differences in agricultural productivity in the two regions. Medieval Hampshire, like the rest of Europe, was heavily reliant on rainfed, annual grain crops, like wheat and rye (Hamerow et al., 2025). This agricultural system was characterized by extensive field systems, and so the societies were land and energy constrained—more land was required per (and between) settlements to support the population. In contrast, south and southeast Asian societies, like Angkor, relied heavily on wet rice paddy cultivation (Klassen et al., 2022). Rice cultivation, particularly with irrigation support, is land-use intensive and highly productive. It can also produce multiple crops per year, something many scholars suspect was happening at Angkor, massively increasing the total annual output (Fox & Ledgerwood, 1999). In fact, estimates of seventeenth-century dry weight productivity of grain and rice crops suggest that the latter produced 2–5× greater dry weight staple crop yield than the former (Malanima, 2006). If true, this would mean that Angkor required half the land area to support the same population, which would correspond rather well to the scaling parameter we estimated—Angkorian communities could have packed in closer together simply because they did not need as much land as Late Anglo-Saxons to support their population.

These potential explanations raise the possibility that scaling differences may reflect fundamental factors such as modal transportation speeds, climate, agricultural land-use systems, and/or terrain. Testing this hypothesis across additional cases, and by employing land-use models, could reveal whether such environmental and infrastructural constraints systematically shape rural–urban scaling patterns, potentially explaining the observed similarities and differences between the point patterns we investigated.

Urbanism has long been a structuring principle of human interaction, and it will only become more central to shaping human life as the share of people living in cities increases. In order to understand urban processes and predict urban futures, we need to include the long history of urbanism from around the world, and we need to improve our understanding of the variability in urban forms and historical trajectories. Our study indicates that now-normative views about premodern urban processes and forms may be missing an underlying shared constraint on urbanism that has in the past affected, and may in the future affect, its local expression. Importantly, this means that the existing dichotomy between the compact/nucleated and low-density agro-urban models may be more a function of perception than ontology.

In the cases we tested, the two apparent “types” (low-density and not) may actually reflect positions on a latent continuum. Our findings also align with the growing body of research from scaling theory, suggesting that urbanism is a recurring outcome of collective human behaviour at scale with scrutable, shared tendencies, shaped undoubtedly by local social/cultural/economic/technological conditions. As both a product and a driver of social, political, and economic organization, it is crucial that we refine our understanding of its true nature, structure, development, and variability. Doing so will not only deepen our understanding of human behaviour at scale but also improve our ability to craft sustainable policies that align with the fundamental processes shaping urban life.

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Author Contribution All authors conceived of the study; W.C.C. wrote the software, collected the data, performed all analyses, and led the writing of the manuscript; all authors reviewed and revised the manuscript.

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Data Availability Data and software used for this study are available at <https://github.com/wccarleton/angkorclusters> and archived at Zenodo (<https://doi.org/10.5281/zenodo.15371728>).

Declarations

Competing interests The authors declare no competing interests.

Classification Archaeology; Urban Archaeology.

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